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Inventor(s)	Murray; Calum et al.

Generating a compliance report of data processing activity

Abstract

Certain aspects of the present disclosure provide techniques for providing a compliance report of data processing to a governing authority. In order to adhere to a regulation of a governing authority, upon receiving the request for a compliance report, a data category and each processing capability category is extracted from a live data catalog service. Based on the extracted categories, a record of data processing is generated for each processing capability category associated with a data category. Further, based on the data category extracted, a compliance report template is retrieved. With the compliance report template and records of data processing, a compliance report is generated and provided to the governing authority.

Inventors: Murray; Calum (Fort Collins, CO), Wilkosz; Jacqueline (San Jose, CA),
Feinstein; Victor (Emerald Hills, CA), McInerney; Denise (Redwood City, CA)

Applicant: INTUIT INC. (Mountain View, CA)

Family ID: 83149832

Assignee: INTUIT INC. (Mountain View, CA)

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Primary Examiner: Thiaw; Catherine

Assistant Examiner: Woldemariam; Nega

Attorney, Agent or Firm: Patterson + Sheridan, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application is a continuation of co-pending U.S. patent application Ser. No. 16/459,839, filed Jul. 2, 2019, the contents of which are incorporated herein by reference in their entirety.

INTRODUCTION

(1) Aspects of the present disclosure relate to a method and system of generating compliance reports of data processing activities consistent with regulations from a governing authority. In particular, embodiments of the present disclosure relate to identifying a type of personal data and how the personal data is processed to generate a compliance report upon request of the governing

authority.

BACKGROUND

(2) Organizations are responsible for collecting, storing, and processing a variety of types of personal data for customers, employees, and clients. For example, an organization can be responsible for contact information (e.g., addresses, phone numbers, and email addresses), personal identifying information (e.g., Social Security numbers, Personal Identification Numbers), financial information (e.g., bank account numbers, transactions), tax information (e.g., tax return, marital status, number of dependents), and medical information (e.g., current medication, history of illnesses, family history). While organizations do take steps to protect the personal data within their purview, including encryption and limiting access to personal data, not all organizations protect personal data in the same manner or to the same extent. Further, the protection of personal data is even more difficult for organizations that process personal data in a distributed computing environment in which multiple computing systems of an organization work together and process personal data.

(3) Governing authorities on a global scale have stepped in and passed laws and issued regulations aimed toward protecting the privacy of and securing personal data collected, processed, and/or stored by organizations. Governing authorities (e.g., local governments, national governments, industry authorities) require compliance with the data protection laws and regulations and can request written compliance reports as proof of the compliance at any time. In particular, the European Union passed in 2016 the General Data Protection Regulation (GDPR) to ensure that personal data collected by organizations is protected within the European Union. Article 30 of the GDPR requires that organizations generate and maintain compliance reports about an organization's personal data processing activities including what personal data is collected, the purpose of collecting the personal data, how long the personal data will be held, and who the personal data is sent to. Additionally, Article 30 requires the compliance reports contain specific pieces of information.

(4) Traditionally, compliance reports have been manually generated in order to meet the requirements of the GDPR. However, manual generation of compliance reports is arduous, time consuming, and subject to human error. Therefore, a solution is needed to provide compliance reports to governing authorities, such as the European Union, without monopolizing the resources of an organization.

BRIEF SUMMARY

(5) Certain embodiments provide a method for providing a compliance report. The method generally includes receiving a request from a requesting system for an electronic compliance report of data processing activity for personal data in a distributed computing environment. The method further includes extracting, based on the request, from a live data catalog service and corresponding to the personal data: a first data category from a set of data categories and each processing capability category associated with the first data category from a set of processing capability categories. The method further includes generating for each processing capability, based on the first data category, a record of data processing activity for the personal data. The method further includes retrieving a compliance report template based on the first data category from a template database. The method further includes generating, based on the compliance report template and each record of data processing activity, the electronic compliance report of the data processing activity in the distributed computing environment for the personal data.

(6) Other embodiments provide systems configured to perform methods for providing a compliance report, such as the aforementioned method, as well as non-transitory computer-readable storage mediums comprising instructions that, when executed by a processor of a processing system, cause the processing system to perform methods for providing a compliance report.

(7) The following description and the related drawings set forth in detail certain illustrative features of one or more embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The appended figures depict certain aspects of the one or more embodiments and are therefore not to be considered limiting of the scope of this disclosure.
- (2) FIG. 1 depicts an example computing environment for generating a compliance report consistent with regulations of a governing authority.
- (3) FIG. 2 depicts an example flow diagram of the generation of a compliance report consistent with regulations of a governing authority.
- (4) FIG. 3 depicts an example method for generating a compliance report consistent with the regulations of a governing authority.
- (5) FIG. 4 depicts an example generated compliance report.
- (6) FIG. 5 depicts an example compliance report system for generating a compliance report consistent with regulations of a governing authority.
- (7) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the drawings. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

- (8) Aspects of the present disclosure provide apparatuses, methods, processing systems, and computer readable mediums for generating compliance reports consistent with regulations from a governing authority using a live data catalog service and compliance report templates for data processed by an organization in a distributed computing environment.
- (9) To adhere to the regulations of a governing authority, upon receiving a request from a requesting system of the governing authority for a compliance report, a compliance report system of an organization identifies a type of personal data and how that type of personal data is processed in the distributed computing environment from a live data catalog service. The live data catalog service maintains an up-to-date record of where and how data is processed by the organization's distributed computing environment, including which computing systems of the organization are handling what type of data and how the data is being processed by the computing systems. The compliance report system then generates a record describing how the personal data is processed, which in turn is used to generate a compliance report, based on a compliance report template, for the governing authority.
- (10) In one embodiment, an organization processes personal data in a distributed computing environment using multiple computing systems. In such instance, a compliance report system of an organization can receive a request to produce a compliance report for a record of how personal data is processed by the organization, such as a request for an electronic compliance report. In some cases, the request for a compliance report may be for each data category and how each data category is processed. In other cases, the compliance report request may be for specific data categories and how each is processed. For example, the European Union can request an electronic compliance report from an organization operating within the European Union as part of Article 30 compliance of the General Data Protection Regulation (GDPR). As part of the request, the European Union can request a separate electronic compliance report for each data category and how each data category is processed by the organization.
- (11) Based on the request, the compliance report system extracts a data category and each processing capability applied to the data category from a live data catalog service. The live data catalog service maintains a record of where and how data is processed in a distributed computing environment of an organization. In some cases, the live data catalog service maintains an up-to-date record of data processing by the organization as well as a historical record of data processing by the

organization. For example, the data in a live data catalog service is updated manually or automatically updated through an API. In some cases, the live data catalog service is a user interface over a database storing data categories and processing capabilities.

(12) The data category indicates a type of personal data, and the processing capability indicates how the personal data is processed. For example, the organization can collect contact information from customers and use the contact information of customers for accounting and small business reporting. In such example, the data category is contact information, and the processing capabilities are accounting and small business reporting. In some cases, the data categories can be created based on domain knowledge.

(13) Using the type of data category extracted from the live data catalog service, the compliance report system retrieves a compliance report template. In some cases, the compliance report template may be consistent with data protection rule(s) established by the regulatory system requesting the compliance report. For example, the compliance report template may be consistent with a data protection rule defined by Article 30 of the GDPR established by the European Union.

(14) Further, using the data category and one of the processing capabilities, the compliance report system generates a record of data processing activity. A record is generated for each processing capability applied to a data category, indicating how data is processed by the organization. With the compliance report template and record of data processing, a compliance report is generated consistent with the request from the governing authority. For example, the electronic compliance report generated may fulfill the requirements of Article 30 of the GDPR.

(15) The method of generating a compliance report consistent with the regulations of a governing authority allows for the compliance report system of an organization to provide compliance reports on-demand. Further, the method of generating the compliance report reduces the amount of resources the organization needs to devote to compliance reporting with the governing authority, and in turn, allows the organization to utilize resources elsewhere, including further data protection. For example, rather than spending time manually generating reports, the organization can now allocate the time and resources saved by generating the compliance report with a template and live data catalog service to develop and implement new methods of data protection.

(16) In addition, an organization processing large amounts of different data types in a distributed computing environment cannot feasibly generate compliance reports on-demand without a compliance report system that utilizes a template database and live data catalog service to generate a compliance report. The template database can include all of the templates for generating compliance reports that have been formatted and defined in such a manner that the templates adhere to compliance reporting regulations. The live data catalog service can maintain an up-to-date record (as well as a historical record) of data processing in the distributed computing environment including the data category associated with data and the processing capabilities applied to the data. With the template database and live data catalog service, the compliance report system can accurately report data processing activities in real time or approximately real time of a rapidly changing distributed computing environment. Additionally, the template database and live data catalog service can be leveraged by the compliance report system to keep up with changing data protection laws.

(17) Without a template database and live data catalog service, the compliance report system would have to search through all of the computing systems of a distributed computing environment to determine which computing systems process what types of data and how, which can be a lengthy and computationally intensive process. Instead, the compliance report system utilizing a template database and live data catalog service can query the live data catalog service and generate a compliance report with data extracted from the live data catalog service using a template from the template database, thereby reducing the amount of time and computation resources needed for generating a compliance report.

Example Computing Environment for Generating a Compliance Report

(18) FIG. 1 depicts an example computing environment **100** for generating a compliance report consistent with regulations of a governing authority. The example computing environment **100** comprises a requesting system **102**, a compliance report system **106**, a live data catalog service **108**, and a template database **116**.

(19) As depicted, the compliance report system **106** receives a request for a compliance report from a requesting system **102**. In some cases, the compliance report system **106** receives a request for an electronic compliance report. The compliance report system **106** may be associated with an organization that collects and/or processes personal data from employees, customers, or clients, or that of others. In some cases, the organization collects and/or processes personal data in a distributed computing environment. As such, the compliance report system **106** may be associated with a live data catalog service **108** that organizes the personal data collected and/or processed by the organization. In some implementations, the live data catalog service **108** organizes the personal data collected and/or processed by a type of data category **112**. For example, data categories **112** can include accounts, billing, documents, financial transaction, and contact information, to name a few, or other information relevant to an individual or to a transactional relationship. In the live data catalog service **108**, each data category **112** is assigned one or more processing capability categories **110**, such as accounting and document management, indicating how the data is processed. Further, the live data catalog service **108** maintains an up-to-date record of where and how data is collected and/or processed by the distributed computing environment of the organization. In some cases, the live data catalog service **108** can maintain a historical record of where and how data is collected and/or processed by the organization.

(20) The requesting system **102** may be associated with a governing authority that has established regulations for protection of the personal data and requires compliance reports as proof of compliance. In some cases, the requesting system **102** is associated with a governing authority such as the European Union, which has established the General Data Protection Regulation (GDPR) for protecting personal data of individuals collected and/or processed by an organization in the European Union.

(21) In other cases, the requesting system **102** is associated with a governing authority such as the United States Department of Health and Human Services, which is responsible for establishing regulations according to the Health Insurance Portability and Accountability Act of 1996 (HIPAA) to secure and protect health information of patients in the United States. The United States Department of Health and Human Services may require organizations with medical information of patients to provide how the medical information is processed to determine whether such organizations are protecting the medical information of patients as required by HIPAA. The requesting system **102** is not limited to governmental organizations and can include industry organizations that have established regulatory authority.

(22) Upon receiving the request from the requesting system **102**, the compliance report system **106** generates a compliance report (e.g., an electronic compliance report) consistent with the regulation of the requesting system **102**. In some cases, the compliance report system **106** can generate a compliance report consistent with the requirements of Article 30 of the GDPR. In other cases, the compliance report system **106** can generate a compliance report consistent with the requirements of HIPAA. As depicted, the generated compliance report is provided to the requesting system **102**. In some implementations, the generated compliance report is provided to the requesting system **102** on-demand. In other implementations, the generated compliance report is provided at a specific time requested by the requesting system **102**. For example, the requesting system **102** can request that the compliance report system **106** provide the compliance report in 24 hours.

(23) The compliance report system **106** that generates the compliance report includes a request module **104**, a data record module **120**, a template module **114**, and a compliance report module **118**.

(24) As depicted, the request module **104** of the compliance report system **106** receives the request

from the requesting system **102** for a compliance report. For example, the request module **104** can receive a request from the European Union, under Article 30 for a compliance report indicating what personal data the organization has and how the personal data is processed. In some cases, the request received by the compliance report system **106** via the request module **104** may be for a portion of the personal data. For example, the requesting system **102** can request a compliance report for one data category of personal data. In other cases, the request may be for a compliance report for each data category of personal data collected and/or processed by the organization.

(25) After receiving the request for a compliance report, the request module **104** sends a signal to a data record module **120** to generate a record of data processing indicating how data is processed. For example, the record of data processing comprises a data category and a corresponding processing capability category. Further, the record of data processing can include which individual (e.g., customer, client, employee) the personal data belongs to and which (if any) third party receives the personal data from the organization.

(26) As depicted, the data record module **120** receives the signal to generate a data record from the request module **104** and extracts a data category **112** and each corresponding processing capability category **110** indexed in the live data catalog service **108**. In some implementations, the signal from the request module **104** may be to generate a data record for a portion of personal data. In such cases, the data record module **120** determines which data category **112** corresponds to the portion of personal data. Based on the determination, the data category **112** and corresponding processing capability categories **110** are extracted from the live data catalog service **108**. In other implementations, the signal from the request module **104** may be to generate data records for all of the personal data. In such cases, the data record module **120** determines each data category for the organization. For example, the data record module **120** can determine there are ten data categories **112** established in the live data catalog service **108**. Based on the determination, each data category **112** and corresponding processing capability categories **110** are extracted from the live data catalog service **108**.

(27) The data record module **120** utilizes the extracted data category **112** and corresponding processing capability category **110** to generate a record of data processing for each processing capability category. In some implementations, a single processing capability category **110** corresponds to a data category **112**. In such cases, the data record module **120** generates one data processing record. In other implementations, multiple processing capability categories **110** correspond to a single data category **112**. In such cases, multiple data processing records are generated. In some implementations, the data record module **120** generates data records for each data category **112** extracted from the live data catalog service **108**.

(28) After the data processing records are generated for each processing capability category **110** associated with a data category **112**, the data record module **120** sends a signal to the template module **114** to retrieve a compliance report template from a template database **116**. In some implementations, the signal indicates the number of data categories **112** extracted from the live data catalog service **108**. In such cases, the template module **114** retrieves a compliance report template for each data category **112**. In some implementations, the template module **114** retrieves the compliance template report from a template database **116** that comprises compliance report templates formatted specific to a regulation. For example, the compliance report templates can be in a format that adheres to the requirements of Article 30 of the GDPR. In some implementations, the template database **116** is co-located with the compliance report system **106**. In other implementations, the template database **116** is located remotely to the compliance report system **106**.

(29) The retrieved compliance report template is sent to the compliance report module **118** by the template module **114**. Further, the compliance report module **118** receives the records of data processing generated from the data record module **120**. With the compliance report template and the records of data processing, the compliance report module **118** generates a compliance report

indicating how personal data is processed by the organization. In some implementations, the compliance report module **118** can generate a separate compliance report document for each data category **112**, as required by the data protection rule(s), based on a single compliance report template retrieved by a template module **114**. For example, the same compliance report template can be used to generate multiple compliance reports. In other implementations, multiple compliance report templates are retrieved and each compliance report template is used to generate a separate compliance report for each data category **112**. In still other implementations, the compliance report module **118** generates a single compliance report for all of the data categories **112**.

(30) As depicted, the compliance report module **118** provides the generated compliance report to the requesting system **102**. In some implementations, the generated compliance report can be provided on demand to the requesting system **102**. In other implementations, the generated compliance report can be provided to the requesting system **102** at a time specified by the requesting system **102**. In such cases, prior to providing the generated compliance report to the requesting system, the compliance report can be stored in a database associated with the compliance report system **106**.

(31) Further, in some implementations, prior to sending the compliance report to the requesting system, the generated compliance report can be reviewed by the compliance report system **106**. For example, the generated compliance report can be displayed in a user interface on a computing device of a reviewer associated with the organization. The reviewer can determine whether the compliance report has any errors. If the reviewer determines there is an error, the reviewer can provide feedback via the user interface, indicating the error. If there is no error, the reviewer can indicate that the generated compliance report is ready to be sent to the requesting system **102**.

(32) While not depicted, the compliance report system **106** may interact with the requesting system **102**, live data catalog service **108**, and template database **116** via a network, which allows the compliance report system **106** to receive the request for the compliance report, extract a data category **112** and corresponding processing capability categories **110**, retrieve a compliance report template, and provide the generated compliance report. The network can include, for example, local area networks (LANs), wireless LANs, wide area networks (WANs), wireless WANs (WWANs), 3G, 4G, LTE, and other technologies capable of transmitting data between devices.

Example Flow Diagram for Generation of a Compliance Report

(33) FIG. 2 depicts an example flow diagram **200** for generation of a compliance report consistent with regulations of a governing authority, as described with respect to FIG. 1.

(34) The example flow diagram **200** begins with the compliance report system **106** receiving **202** a request for an electronic compliance report of data processing activity from a requesting system **102**. Upon receiving the request, the compliance report system **106** requests **204** and receives **206** a data category and each processing capability category associated with the data category from a live data catalog service **108** based on the request. The compliance report system **106** generates **208** a record of data processing for each processing capability category of a data category.

(35) Once each record of data processing is generated for each processing capability applied to a data category, the compliance report system **106** requests **210** and receives **212** a compliance report template from a template database **116**. In some implementations, the compliance report system **106** retrieves a compliance report template based on the data category extracted. In some cases, a separate template is retrieved for each data category. For example, different data categories can have different compliance reporting requirements. In other cases, a single compliance report template is retrieved.

(36) With the compliance report template and the records of data processing, the compliance report system **106** generates **214** an electronic compliance report. The compliance report system **106** then provides **216** the generated electronic compliance report to the requesting system **102**.

Example Method for Generating a Compliance Report

(37) FIG. 3 depicts an example method **300** for generating a compliance report consistent with regulations of a governing authority, as described with respect to FIGS. 1-2.

(38) At step **302**, a request is received for a compliance report of data processing activity for personal data. In some cases, the request for an electronic compliance report is received by an organization under a regulation from a requesting system associated with a governing authority. For example, the European Union can request an electronic compliance report under Article 30 of the GDPR.

(39) At step **304**, a data category and each processing capability associated with the category is extracted from a live data catalog service. Based on the request received from the requesting system of the governing authority, a data category from a set of categories is identified and extracted from a live data catalog service. Further, each processing capability associated with the data category is extracted from the live data catalog service. The extracted data category indicates a type of personal data and processing capabilities indicate how a particular type of personal data is processed by a data process of the organization.

(40) At step **306**, a compliance template report is retrieved. Based on the data category extracted from the live data catalog service, a compliance report template is retrieved from a template database. The compliance report template is formatted to comply with data protection rule(s) established by the governing authority. For example, the compliance report template retrieved from the template database is formatted consistent with data protection rules defined by Article 30 of the GDPR.

(41) At step **308**, a record of data processing activity is generated for each processing capability, based on the data category. The record of data processing indicates how a type of personal data is processed.

(42) At step **310**, a compliance report of the data processing activity for personal data is generated based on the compliance report template and each record of data processing activity. In some cases, a separate electronic compliance report is generated for each data category according to the data protection rule(s). For example, Article 30 of the GDPR can require a separate electronic compliance report for each record of data processing activity. In other cases, a single electronic compliance report comprising all of the generated records of data processing activity is required according to the data protection rule(s). For example, the HIPPA can require a single electronic compliance report that includes all of the records of data processing.

(43) At step **312**, the generated compliance report is sent by the compliance reporting system to the requesting system. In some cases, the generated electronic compliance report is sent to the requesting system upon generation of the electronic compliance report. In other implementations, the generated electronic compliance report is sent to the requesting system at a time requested by the requesting system. For example, the requesting system in the request for the electronic compliance report can request the generated electronic report be sent twenty-four hours after the request is received by the compliance reporting system.

(44) In some cases, the compliance report is sent to a reviewer prior to sending the compliance report to the governing authority. The reviewer is associated with the organization and may review the compliance report on a computing device to determine whether the compliance report has any errors, as described in FIG. 1. If the reviewer determines there is an error with the generated compliance report, the reviewer provides such feedback, and an updated compliance report is generated that corrects the error in the previously generated compliance report.

Example of a Generated Compliance Report

(45) FIG. 4 depicts an example compliance report **400** generated by a compliance report system, as described with respect to FIGS. 1-3.

(46) The compliance report includes data processing records generated for each type of data processing capability applied to a category of personal data. As depicted, the compliance report indicates how general information associated with customers is processed for accounting, tax

reporting, and bank account purposes. Further, the compliance report indicates recipients of the general information of customers.

(47) In some implementations, the generated compliance report is displayed by a computing device associated with the compliance report system prior to providing the compliance report. The displayed compliance report allows users associated with the compliance report system of an organization to provide feedback via the display. For example, if a user identifies an error in the generated compliance report, then the user may indicate the inaccuracy via the display of the computing device and provide feedback indicating what the error is. In some cases, the feedback can be used to regenerate a correct compliance report. Further, if the user determines there are no errors, then the user can indicate the compliance report is ready to be submitted to the requesting system.

(48) In other implementations, the generated compliance report is displayed to a requesting system of the governing authority to determine compliance with regulation.

Example of a Compliance Report System for Generating a Compliance Report

(49) FIG. 5 depicts an example of a compliance report system **500** that may be used to perform methods described herein, such as the method for generating a compliance report consistent with regulations of a governing authority described with respect to FIGS. 1-3.

(50) Compliance report system **500** includes a central processing unit (CPU) **502** connected to a data bus **508**. CPU **502** is configured to process computer-executable instructions, e.g., stored in memory **510** or storage **512**, and to cause the compliance report system **500** to perform methods described herein, for example with respect to FIGS. 1-3. CPU **502** is included to be representative of a single CPU, multiple CPUs, a single CPU having multiple processing cores, and other forms of processing architecture capable of executing computer-executable instructions.

(51) Compliance report system **500** further includes input/output (I/O) device(s) **514** and interface(s) **504**, which allows compliance report system **500** to interface with input/output devices **514**, such as, for example, keyboards, displays, mouse devices, pen input, and other devices that allow for interaction with compliance report system **500**. Note that while not depicted with independent external I/O devices, compliance report system **500** may connect with external I/O devices through physical and wireless connections (e.g., an external display device).

(52) Compliance report system **500** further includes network interface **506**, which provides compliance report system **500** with access to external networks **516** and thereby external computing devices.

(53) Compliance report system **500** further includes memory **510**, which in this example includes request module **104** that receives a request from a governing authority for a compliance report and generates a request for a record of data processing based on the request for the compliance report, as described above, for example with respect to FIGS. 1-2.

(54) Memory **510** further includes data record module **120** that generates a record of data processing by extracting a data category and each processing capability applied to the data category from a live data catalog service, as described above, for example with respect to FIGS. 1-2.

(55) Memory **510** further includes template module **114** that retrieves a compliance report template based on the data category extracted from the live data catalog service, as described above, for example with respect to FIGS. 1-2.

(56) Memory **510** further includes compliance report module **118** that generates a compliance report based on the compliance report template and the record of data processing, as described above, for example with respect to FIGS. 1-2.

(57) Note that while shown as a single memory **510** in FIG. 5 for simplicity, the various aspects stored in memory **510** may be stored in different physical memories, but all accessible by CPU **502** via internal data connections such as bus **508**.

(58) Compliance report system **500** further includes storage **512**, which in this example includes request data **518**, which may be received from a requesting system of a governing authority for a

compliance report, as described with respect to FIGS. 1-3.

(59) Storage **512** further includes data categories **520**, which may be like a data category **112**, as described with respect to FIG. 1.

(60) Storage **512** further includes processing capability categories **522**, which may be like a processing capability category **110**, as described with respect to FIG. 1.

(61) Storage **512** further includes compliance templates **524**, which may be like a compliance report template, as described with respect to FIGS. 1-3.

(62) Storage **512** further includes records of data processing **526**, which may be like a record of data processing, as described with respect to FIGS. 1-3.

(63) Storage **512** further includes compliance reports **528**, which may be like a compliance report, as described with respect to FIGS. 1-3.

(64) While not depicted in FIG. 5, other aspects may be included in storage **512**.

(65) As with memory **510**, a single storage **512** is depicted in FIG. 5 for simplicity, but the various aspects stored in storage **512** may be stored in different physical storages, but all accessible to CPU **502** via internal data connections, such as bus **508**, or external connection, such as network interface **506**.

(66) The preceding description is provided to enable any person skilled in the art to practice the various embodiments described herein. The examples discussed herein are not limiting of the scope, applicability, or embodiments set forth in the claims. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments. For example, changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various steps may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

(67) As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

(68) As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” may include resolving, selecting, choosing, establishing and the like.

(69) The methods disclosed herein comprise one or more steps or actions for achieving the methods. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor. Generally, where there are

operations illustrated in figures, those operations may have corresponding counterpart means-plus-function components with similar numbering.

(70) The various illustrative logical blocks, modules and circuits described in connection with the present disclosure may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device (PLD), discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any commercially available processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

(71) A processing system may be implemented with a bus architecture. The bus may include any number of interconnecting buses and bridges depending on the specific application of the processing system and the overall design constraints. The bus may link together various circuits including a processor, machine-readable media, and input/output devices, among others. A user interface (e.g., keypad, display, mouse, joystick, etc.) may also be connected to the bus. The bus may also link various other circuits such as timing sources, peripherals, voltage regulators, power management circuits, and other circuit elements that are well known in the art, and therefore, will not be described any further. The processor may be implemented with one or more general-purpose and/or special-purpose processors. Examples include microprocessors, microcontrollers, DSP processors, and other circuitry that can execute software. Those skilled in the art will recognize how best to implement the described functionality for the processing system depending on the particular application and the overall design constraints imposed on the overall system.

(72) If implemented in software, the functions may be stored or transmitted over as one or more instructions or code on a computer-readable medium. Software shall be construed broadly to mean instructions, data, or any combination thereof, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. Computer-readable media include both computer storage media and communication media, such as any medium that facilitates transfer of a computer program from one place to another. The processor may be responsible for managing the bus and general processing, including the execution of software modules stored on the computer-readable storage media. A computer-readable storage medium may be coupled to a processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. By way of example, the computer-readable media may include a transmission line, a carrier wave modulated by data, and/or a computer readable storage medium with instructions stored thereon separate from the wireless node, all of which may be accessed by the processor through the bus interface. Alternatively, or in addition, the computer-readable media, or any portion thereof, may be integrated into the processor, such as the case may be with cache and/or general register files. Examples of machine-readable storage media may include, by way of example, RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, magnetic disks, optical disks, hard drives, or any other suitable storage medium, or any combination thereof. The machine-readable media may be embodied in a computer-program product.

(73) A software module may comprise a single instruction, or many instructions, and may be distributed over several different code segments, among different programs, and across multiple storage media. The computer-readable media may comprise a number of software modules. The software modules include instructions that, when executed by an apparatus such as a processor, cause the processing system to perform various functions. The software modules may include a

transmission module and a receiving module. Each software module may reside in a single storage device or be distributed across multiple storage devices. By way of example, a software module may be loaded into RAM from a hard drive when a triggering event occurs. During execution of the software module, the processor may load some of the instructions into cache to increase access speed. One or more cache lines may then be loaded into a general register file for execution by the processor. When referring to the functionality of a software module, it will be understood that such functionality is implemented by the processor when executing instructions from that software module.

(74) The following claims are not intended to be limited to the embodiments shown herein, but are to be accorded the full scope consistent with the language of the claims. Within a claim, reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.” All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. A method, comprising: receiving a request from a requesting system for an electronic compliance report of data processing activity for personal data in a computing environment; extracting, based on the request, from a live data catalog service and corresponding to the personal data: a first data category from a set of data categories; and a processing capability category associated with the first data category from a set of processing capability categories, wherein the live data catalog service maintains a record of operations in which data corresponding to the set of data categories is processed according to the set of processing capability categories in the computing environment; and generating for the processing capability category, based on the first data category, a record of data processing activity for the personal data; retrieving a compliance report template based on the first data category from a template database; and generating, based on the compliance report template and the record of data processing activity, the electronic compliance report of the data processing activity in the computing environment for the personal data.
2. The method of claim 1, further comprising sending the generated electronic compliance report to the requesting system.
3. The method of claim 1, further comprising: sending the generated electronic compliance report to a reviewer; receiving feedback from the reviewer of an error in the generated electronic compliance report; and generating an updated electronic compliance report to address the error.
4. The method of claim 1, wherein the retrieved compliance report template is formatted to comply with a data protection rule.
5. The method of claim 4, wherein the data protection rule is defined by General Data Protection Regulation, Article 30.
6. The method of claim 5, further comprising generating a separate electronic compliance report for each data category in the set of data categories according to the data protection rule.
7. The method of claim 1, wherein the record of data processing activity includes at least: the first data category indicating a type of the personal data; and the processing capability category indicating a processing capability applied to the personal data.
8. A method, comprising: receiving a request from a requesting system for an electronic compliance report of data processing activity for data in a computing environment; extracting, based on the

request, from a live data catalog service and corresponding to the data: a first data category from a set of data categories; and a processing capability category associated with the first data category from a set of processing capability categories, wherein the live data catalog service maintains a record of operations in which data corresponding to the set of data categories is processed according to the set of processing capability categories in the computing environment; and generating for the processing capability category, based on the first data category, a record of data processing activity for the data; retrieving a compliance report template based on the first data category from a template database; and generating, based on the compliance report template and the record of data processing activity, the electronic compliance report of the data processing activity in the computing environment for the data.

9. The method of claim 8, further comprising sending the generated electronic compliance report to the requesting system.

10. The method of claim 8, further comprising: sending the generated electronic compliance report to a reviewer; receiving feedback from the reviewer of an error in the generated electronic compliance report; and generating an updated electronic compliance report to address the error.

11. The method of claim 8, wherein the retrieved compliance report template is formatted to comply with a data protection rule.

12. The method of claim 11, wherein the data protection rule is defined by General Data Protection Regulation, Article 30.

13. The method of claim 12, further comprising generating a separate electronic compliance report for each data category in the set of data categories according to the data protection rule.

14. The method of claim 8, wherein the record of data processing activity includes at least: the first data category indicating a type of the personal data; and the processing capability category indicating a processing capability applied to the personal data.

15. A system, comprising: one or more processors; and a memory comprising instructions that, when executed by the one or more processors, cause the system to: receive a request from a requesting system for an electronic compliance report of data processing activity for personal data in a computing environment; extract, based on the request, from a live data catalog service and corresponding to the personal data: a first data category from a set of data categories; and a processing capability category associated with the first data category from a set of processing capability categories, wherein the live data catalog service maintains a record of operations in which data corresponding to the set of data categories is processed according to the set of processing capability categories in the computing environment; and generate for the processing capability category, based on the first data category, a record of data processing activity for the personal data; retrieve a compliance report template based on the first data category from a template database; and generate, based on the compliance report template and the record of data processing activity, the electronic compliance report of the data processing activity in the computing environment for the personal data.

16. The system of claim 15, wherein the instructions, when executed by the one or more processors, further cause the system to send the generated electronic compliance report to the requesting system.

17. The system of claim 15, wherein the instructions, when executed by the one or more processors, further cause the system to: send the generated electronic compliance report to a reviewer; receive feedback from the reviewer of an error in the generated electronic compliance report; and generate an updated electronic compliance report to address the error.

18. The system of claim 15, wherein the retrieved compliance report template is formatted to comply with a data protection rule.

19. The system of claim 18, wherein the data protection rule is defined by General Data Protection Regulation, Article 30.

20. The system of claim 19, wherein the instructions, when executed by the one or more

processors, further cause the system to generate a separate electronic compliance report for each data category in the set of data categories according to the data protection rule.
